Math 55 worksheet Wednesday 02/11/2026

1. Using the Euclidean algorithm, show that $\gcd(48, 18) = 6$. Using this, find $x, y \in \mathbb{Z}$ so that $6 = 48x+18y$.

Solution:

$$\begin{aligned} 48 &= 2 \cdot 18 + 12 \\ 18 &= 1 \cdot 12 + 6 \\ 12 &= 2 \cdot 6 + 0 \end{aligned}$$

The last nonzero remainder, in this case 6, is $\gcd(48, 18)$. Solving for the remainders:

$$\begin{aligned} 48 = 2 \cdot 18 + 12 &\to 12 = 48 - 2 \cdot 18 \\ 18 = 1 \cdot 12 + 6 &\to 6 = 18 - 1 \cdot 12 \end{aligned}$$

Starting with the last one and back-substituting:

$$6 = 18 - 1 \cdot 12 = 18 - 1 \cdot (48 - 2 \cdot 18) = 3 \cdot 18 - 1 \cdot 48$$

as desired.

2. Using Bézout's theorem, show if $\gcd(\mathrm{a}, \mathrm{m}) = 1$ then there exists $x \in \mathbb{Z}$ so that $ax \equiv 1 \bmod(\mathrm{m})$.

Solution: By Bézout's theorem, we know there exists $x, y \in \mathbb{Z}$ so that:

$$ax + my = \gcd(a, m) = 1$$

As $my \equiv 0 \bmod(\mathrm{m})$ we see this implies:

$$1 \equiv ax + my \equiv ax \bmod(\mathrm{m}).$$

Hence the desired multiplicative inverse x is exactly the coefficient of a coming from Bézout's theorem.

3. Show that if $n \mid ab$ and $\gcd(n, a) = 1$, namely n and a are coprime, then $n \mid b$.

Solution: The key idea is that Bézout's theorem gives us a way of connecting the greatest common divisor of n, a to n, a themselves, from which we can further manipulate to get what we need. Namely, by Bézout's theorem, we see there exists $x, y \in \mathbb{Z}$ so:

$$1 = nx + ay$$

Multiplying both sides by b:

$$b = nbx + aby$$

Since $n \mid nbx$ and $n \mid aby$ by assumption, we must have $n \mid nbx + aby$. As this equals b, $n \mid b$.

4. Show that if $ac \equiv bc \bmod(\mathrm{m})$ then:

$$a \equiv b \bmod \left(\frac{m}{\gcd(m,c)}\right)$$

Solution:

By assumption, we know there exists $k \in \mathbb{Z}$ so that:

$$km = ac - bc = c(a-b)$$

Let $d = \gcd(m,c)$. Then as $d \mid m$ and $d \mid c$ we can write $m = rd$ and $c = \ell d$ for some $\ell, r \in \mathbb{Z}$. Moreover, as d is the *greatest* common divisor, it must be true that r, ℓ are coprime. Hence the above becomes:

$$krd = \ell d(a-b) \to kr = \ell(a-b)$$

Since $\ell \mid \ell(a-b)$, it must divide kr as well. As ℓ, r are coprime and ℓ divides, the previous question implies $\ell \mid k$. Hence $a - b$

5. Suppose n, m have the following prime factorizations:

$$n = \prod_{i=1}^{k} p_i^{a_i}, \quad m = \prod_{i=1}^{k} p_i^{b_i},$$

Using this, find a formula for $\gcd(\mathrm{n},\mathrm{m})$ and $\mathrm{lcm}(\mathrm{n},\mathrm{m})$ in terms of these prime factors. With this, show $\gcd(\mathrm{n},\mathrm{m}) \cdot \mathrm{lcm}(\mathrm{n},\mathrm{m}) = nm$.

Solution: The desired formulas are:

$$\gcd(n,m) = \prod_{i=1}^{k} p_i^{\min(a_i,b_i)}, \quad \mathrm{lcm}(\mathrm{n},\mathrm{m}) = \prod_{i=1}^{n} p_i^{\max(a_i,b_i)}$$

The first formula follows immediately from the fact that $\gcd(\mathrm{n},\mathrm{m}) \mid n$ and $\gcd(\mathrm{n},\mathrm{m}) \mid m$. The second from the fact $n \mid \mathrm{lcm}(\mathrm{n},\mathrm{m})$ and $m \mid \mathrm{lcm}(\mathrm{n},\mathrm{m})$. Since:

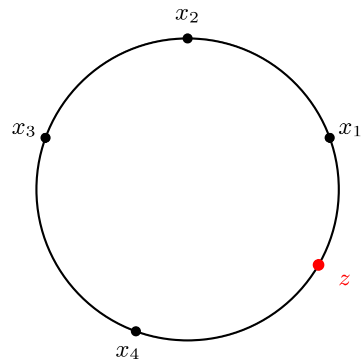
$$\max(x,y) + \min(x,y) = x + y$$

and nm has prime factorization:

$$nm = \prod_{i=1}^{k} p_i^{a_i+b_i}$$

it follows that $\gcd(\mathrm{n},\mathrm{m}) \cdot \mathrm{lcm}(\mathrm{n},\mathrm{m}) = nm$.

6. (Fun, unrelated, and challenging). Suppose we have a circular race track with circumference one. Along the track are n gas cans at angles $x_1, ..., x_n \in [0, 2\pi)$. The can at location x_i contains enough fuel to travel distance $d_i \geq 0$. There is exactly enough gas across the n cans to traverse the track: $\sum_{i=1}^{n} d_i = 1$. We aim to drive a car clockwise along this track starting with no fuel and collecting gas from the gas cans along the way. Prove or disprove: for any choice of $n, \{x_i\}_{i=1}^{n}, \{d_i\}_{i=1}^{n}$ there exists a starting point $z \in [0, 2\pi)$ so that our car can make it all the way around the track without running out of fuel.



Solution: We claim that it is always possible to complete the circuit.

Fix an arbitrary reference point $z \in [0, 1)$. As we traverse the track clockwise starting from z, define the *net fuel function*

$$F(t) := \sum_{i : x_i \in [z,t)} d_i - (t - z),$$

where $t \in [z, z+1]$ is interpreted modulo 2π. Intuitively, $F(t)$ is the amount of fuel the car would have upon reaching position t, if negative values are allowed. The function $F(t)$ decreases with slope -1 between gas cans and jumps upward by d_i at each location x_i. Since

$$F(z+1) = \sum_{i=1}^{n} d_i - 1 = 0 = F(z),$$

the function approaches from the left a global minimum at some point $z^* \in [z, z+1)$. Necessarily, z^* occurs immediately before reaching some gas can. Now start the car at z^* with an empty fuel tank. For any point t reached after z^*, the fuel in the tank equals

$$F(t) - F(z^*) \geq 0,$$

since $F(z^*)$ is the minimum of F. Thus the car never runs out of fuel and completes one full lap around the track.